MSSC 6250: Heart Attack Risk Analysis

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Abstract

In the United States, a person has a heart attack every 40 seconds [1]. Identifying heart attack risk factors is critically important when guiding lifestyle decisions. There is a large surplus of data available that can be used to attribute various predictors of a heart attack. Various machine learning (ML) models have been applied to a data set containing 7011 people worldwide with 25 distinct features with entries has been utilized. The Machine Learning (ML) models include logistic regression, Principal Component Analysis (PCA), Support Vector Machine (SVM), K- Nearest Neighbor (KNN), Random Forest, and clustering. Model performance has been quantified by accuracy and F1- score. The top performing model is a random forest model at a percent accuracy. It shows that ML can be applied to a real-world medical data set showing potential to impact health-related decisions for people around the world.

Keywords: *Heart Attack Risk, Machine Learning, Supervised Learning, Unsupervised Learning*

Introduction

Heart attacks, also known as myocardial infractions remain one of the leading causes of death worldwide [1,3]. In the United States alone, the Centers for Disease Control and Prevention (CDC) reports that someone suffers a heart attack every 40 seconds, resulting in hundreds of thousands of deaths each year. A heart attack occurs when there is a blockage of blood flow to a heart muscle, depriving the tissue of oxygen. The severity of heart attacks range in severity. Despite progress in medicine and health technologies, the risk of heart attacks continues to grow globally. Contributing factors include increasingly sedentary lifestyles, unhealthy diets, rising stress levels, and aging populations. These trends highlight the urgent need for better ways to assess risk, detect warning signs early, and take preventive action.

Traditionally, doctors assess the risk of heart attack using some key indicators such as age, cholesterol levels, blood pressure, and family history. Although these are still important, recent advances in data science allow us to consider a much broader range of factors. Machine learning techniques, in particular, offer the ability to detect complex patterns in data, patterns that may not be visible through traditional methods. Looking at behavioral habits, socioeconomic status, and geographic background in conjunction with medical data, we can get a more complete picture of what contributes to the risk of heart attack.

In this project, we analyze a dataset of 7,010 individuals from around the world. Each record includes 25 features that cover medical history, lifestyle behaviors, and demographic details. These include clinical data such as cholesterol, blood pressure, and heart rate, along with information about diabetes, smoking and drinking habits, exercise routines, diet, sleep, income, and location. The goal is to predict a binary outcome, whether or not the person is at risk of a heart attack.

To explore this, we apply several ML methods: Logistic Regression, Principal Component Analysis (PCA), Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forest, and K-means clustering techniques. These models were selected for their balanced combination of predictive power, interpretability, and flexibility. Our objective is to evaluate which models perform best, identify the most important risk factors, and provide meaningful information that could inform preventive health efforts.

Ultimately, this research aims to show how clinical knowledge and machine learning can work together to improve our understanding and prediction of the risk of heart attacks. Machine learning allows us a way to examine patterns in

large, complicated datasets. By applying machine learning, we hope to support doctors, researchers, and public health officials in making better decisions and helping more people stay heart-healthy. It will allow us to inform physicians on the combination of risk factors that contribute to worse health outcomes in terms of heart attack prevention.

Data

An initial exploratory analysis was conducted to understand the structure, distribution, and relationships within the dataset. The dataset comprises 7,011 observations with 26 variables and offers a rich set of features related to heart health and lifestyle behaviors. It includes individual demographic health metrics such as age, gender, cholesterol levels, blood pressure, heart rate, diabetes status, family history of heart disease, smoking, obesity, and alcohol use. Lifestyle factors are also represented, including exercise frequency, diet quality, stress levels, and sedentary behavior. The dataset further incorporates clinical details like past heart conditions, medication use, and triglyceride levels. Socioeconomic and geographic variables—such as income, country, continent, and hemisphere—are also included, enabling broader contextual analysis. Despite the data set being throughout, containing a wide range of features, not all possible heart attack predictors were included. Other factors like genetics, other cardiovascular diseases, and presence of a prior cardiovascular event were not included.

Numerical features such as age and BMI were examined using histograms and boxplots to explore their distributions across the dataset. A scatterplot of BMI versus age, colored by heart attack risk, was used to visualize the relationship between these two continuous variables and to assess potential clustering or nonlinear patterns between risk groups. Additionally, a boxplot comparing BMI distributions between at-risk and not-at-risk individuals provided a clearer summary of median values and spread by risk category. The distribution of age was further detailed with a histogram showing how the sample is spread across different age groups.

Categorical variables like smoking status, sex, diet, and country were explored using proportion and count plots, highlighting lifestyle and demographic differences and their potential association with heart attack risk across groups.

A correlation heatmap (Figure 3) was also used to detect linear relationships between variables, laying the groundwork for feature selection and model building. The heatmap helped identify any collinear present among the features. Overall, the only features that showed notable correlation were smoking and sex, while the remaining 23 features exhibited very little correlation.

Figure 1 depicts the count of individuals at risk and not at risk for heart attacks across age groups. The age range of the data set is between 18 and 90 years old. The age distribution in this dataset appears approximately uniform across the observed age range, meaning individuals are fairly evenly spread across age groups. The red segments (at risk) appear fairly consistent across ages, suggesting risk is spread across age groups without sharp increases in older populations. The plot also shows the class imbalance present in the data, since there were more observations of people not at risk for a heart attack within the dataset.

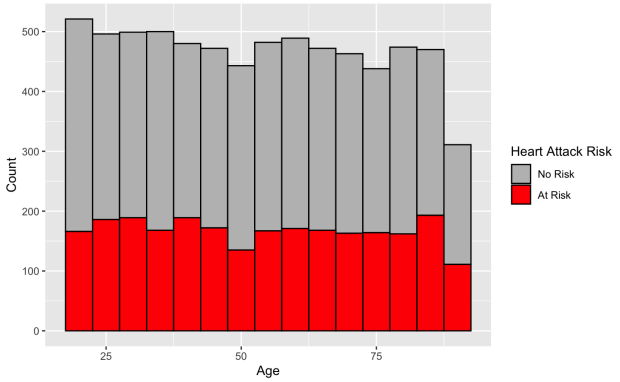

Figure 1. Heart Attack Risk by Age

Figure 2 compares the proportion of heart attack risk between males and females. Both sexes show similar proportions of individuals at risk, indicating no strong difference in heart attack risk between sexes in this dataset. This is an important observation, as sex might otherwise be expected to act as a predictive factor, or certain predictors might apply differently to males and females.

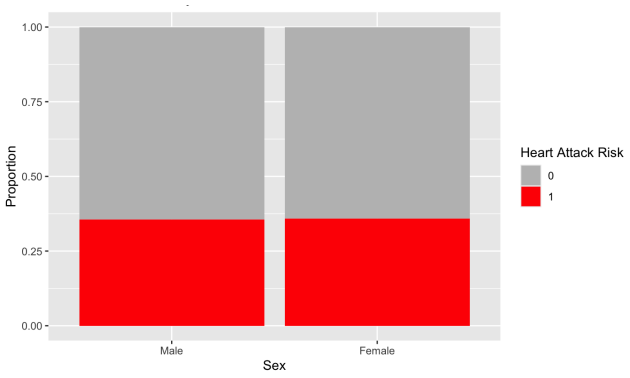

Figure 2. Heart Attack Risk by Sex

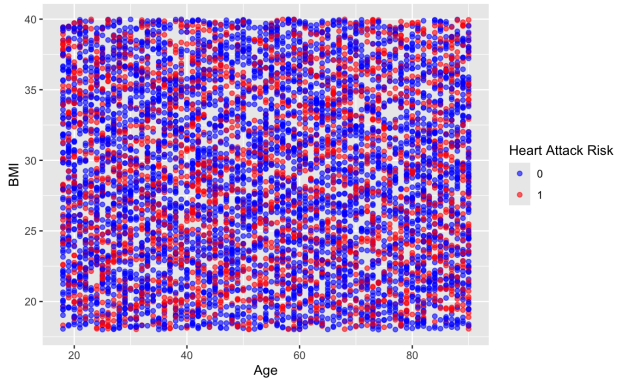

Figure 4. BMI vs Age Colored by Risk

Figure 3 features a boxplot which compares BMI distributions for those at risk versus not at risk for a heart attack. Both groups have similar median BMIs, although the at-risk group shows slightly higher spread, indicating marginally greater BMI variability among those at risk.

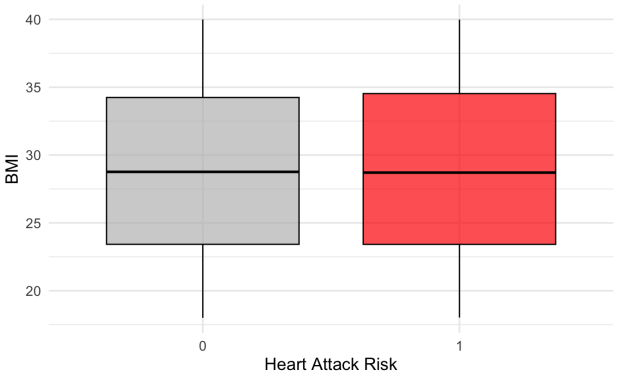

Figure 3. Heart Attack Risk by BMI

Figure 5 shows the proportion of heart attack risk across different countries. While there is some variation, the proportion of individuals at risk remains fairly similar across countries, suggesting heart attack risk is fairly evenly distributed geographically.

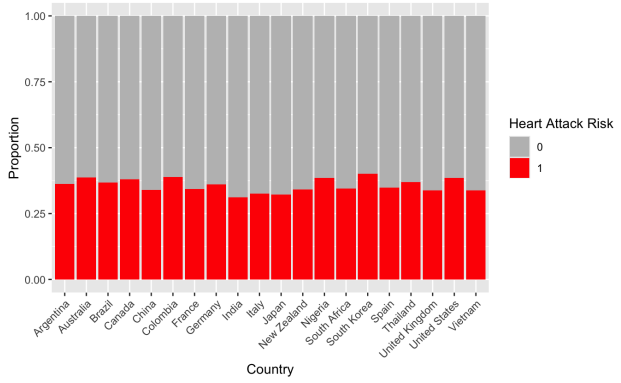

Figure 5. Heart Attack Risk by Country

This scatter plot in Figure 4 displays individuals' BMI and age, colored by heart attack risk. Both at-risk (red) and not-at-risk (blue) individuals are scattered evenly across the BMI-age space, suggesting no simple visual separation by these two variables alone.

The plot in Figure 6 shows the proportion of individuals at risk and not at risk for heart attacks across smoking status (0 = non-smokers, 1 = smokers). The proportions appear nearly identical between smokers and non-smokers, suggesting that in this dataset, smoking status alone does not show a clear difference in heart attack risk.

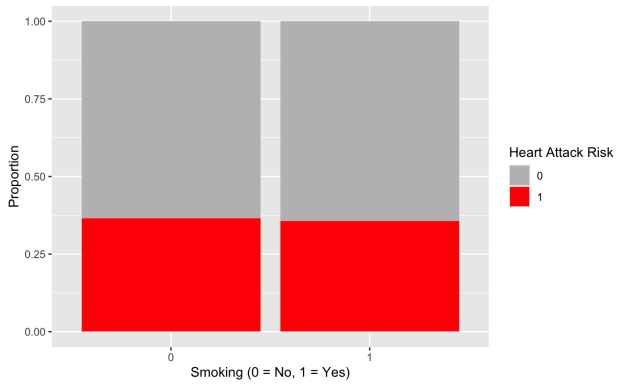

Figure 6. Heart Attack Risk by Smoking Status

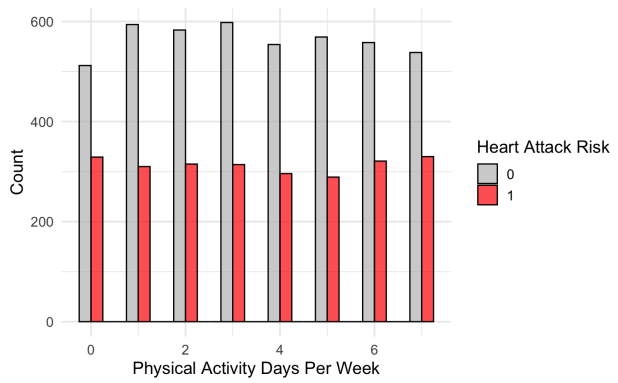

Figure 8. Heart Attack Risk by Weekly Physical Activity

Figure 7 compares the proportion of heart attack risk between individuals with and without diabetes (0 = non-diabetic, 1 = diabetic). While individuals with diabetes show a slightly higher proportion of risk, the overall difference is small, suggesting that diabetes status alone does not strongly differentiate heart attack risk in this dataset.

The plot in Figure 9 shows the proportion of individuals at risk and not at risk for heart attacks based on family history (0 = no family history, 1 = family history present). The proportions are nearly identical between the two groups, suggesting that family history alone does not show a strong difference in heart attack risk within this dataset.

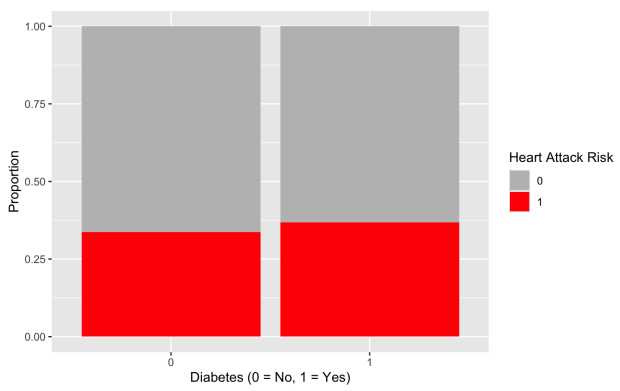

Figure 7. Heart Attack Risk by Diabetes Status

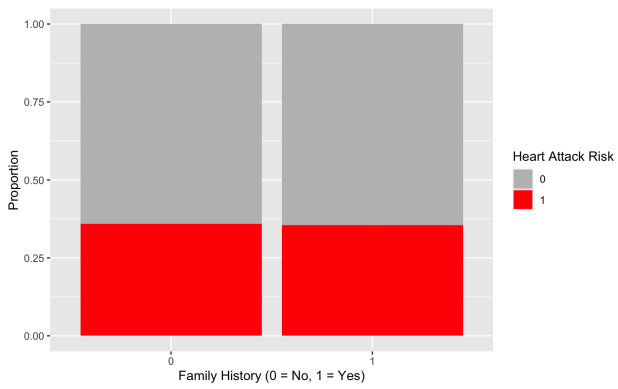

Figure 9. Heart Attack Risk by Family History

The count of individuals at risk and not at risk based on how many days per week they engage in physical activity is shown in Figure 8. Across all activity levels, the counts and proportions of at-risk individuals (in red) are interestingly fairly similar/uniform, indicating that physical activity frequency does not show a strong relationship with heart attack risk in this dataset.

The boxplot in Figure 10 compares cholesterol levels between individuals at risk and not at risk for heart attacks. The distributions appear similar, with slightly higher median cholesterol among at-risk individuals, but overall the spread and range overlap substantially, indicating that cholesterol levels alone may not sharply separate the two risk groups either.

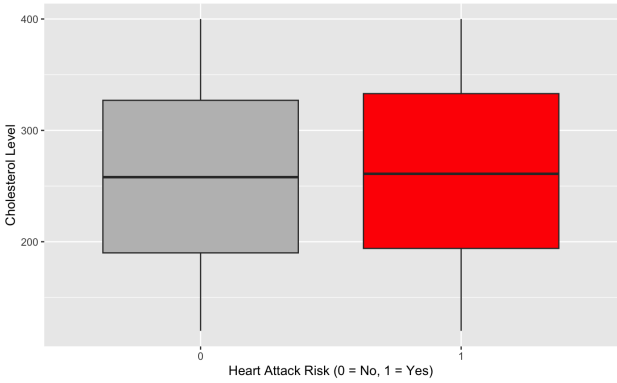

Figure 10. Heart Attack Risk by Cholesterol Levels

Data prepossessing was necessary to ensure the features were in a suitable format to be implemented into the various models. Categorical variables, including sex, diet, country, continent, and hemisphere, were initially stored as character strings and were converted to factors so that the models could process them correctly. For clustering methods such as K-means, as well as models like Support Vector Machines, which require purely numeric input, these categorical factors were further transformed into numeric representations. Blood pressure was also formatted as a character and not in decimal form. To mitigate this, the systolic and diastolic blood pressure was divided into two different features. The last step of preproccesing was to ensure no features had a surplus of missing values. No feature vectors were removed.

Methodology

For our analysis, we chose to use all 25 predictors, and the risk of heart attack is chosen as the response variable. Multiple machine learning algorithms were implemented on the dataset to carry out binary classification. An 80/20 random split was used to divide the data into training and test sets.

To comprehensively evaluate heart attack risk, we applied a range of models, including PCA with logistic regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), K-means clustering, Random Forest, and Naive Bayes. These models were chosen because they represent a diverse set of methodological approaches, covering both linear and nonlinear classification (logistic regression vs. SVM, Random Forest), parametric and non-parametric methods (logistic regression vs. KNN, Random Forest), supervised and unsupervised learning (Random Forest vs. K-means), and probabilistic and distance-based frameworks (Naive Bayes vs. KNN). By incorporating such a variety, we were able to assess which models best captured the patterns in the data and provided the strongest predictive performance.

To address the class imbalance present in the dataset - where the number of individuals at risk of heart attack was substantially lower than those not at risk — we applied the synthetic minority oversampling technique (SMOTE). SMOTE works by generating synthetic samples for the minority class based on nearest neighbors, rather than simply duplicating existing samples. This helps balance the class distribution in the training data, allowing the machine learning algorithms to learn more representative decision boundaries and reducing the bias toward the majority class. Without addressing this imbalance, models may achieve deceptively high accuracy by favoring the majority class while performing poorly at detecting the minority (at-risk) cases, which are of critical interest in this analysis. Given that identifying individuals at risk for heart attacks can directly inform preventive interventions and healthcare decisions, ensuring that the model accurately captures minority class patterns is imperative to reduce the risk of missed diagnoses and to improve public health outcomes.

Figure 11 shows the distribution of the classes before and after SMOTE, including counts and percentages. Notably, after applying SMOTE, 52.6% of the dataset was classified as having no heart attack risk, while 47.4% was classified as at risk. This adjusted distribution closely aligns with real-world estimates reported by the American Heart Association, which noted in 2019 that nearly 48% of adults in the United States suffer from cardiovascular disease and are therefore considered at risk for a heart attack [2].

Class	Before SMOTE	After SMOTE
No Risk (0)	4506 (64.3%)	4506 (52.6%)
At Risk (1)	2504 (35.7%)	4055 (47.4%)

Figure 11. Class Distribution Before and After SMOTE

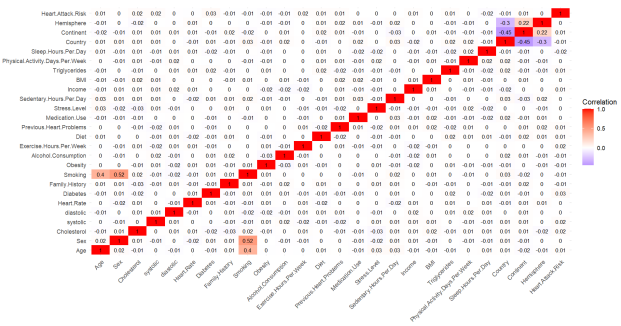

Figure 12. Pearson Correlation between each feature

PCA

Principal Component Analysis (PCA) is a method of dimension reduction that is applied to data sets with a large number of features. The core objective is to explain the variance across all the features in a condensed, lower dimensional way. Variables that are beneficial to a model explain a high amount of variance within a dataset. To do this, the principal components are created. In Figure 13, the two main components are shown. The graph is colored red (no heart attack) and blue (heart attack). Despite the top two principal components creating separation their is extreme overlap between the labeled conditions.

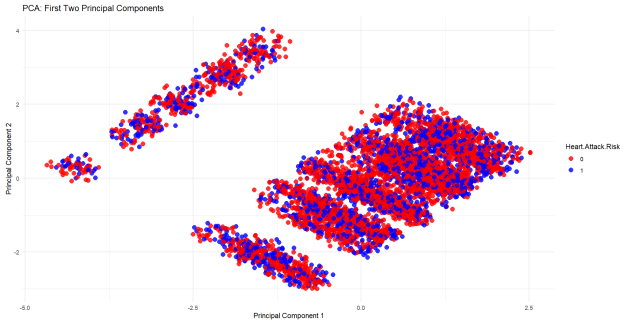

Figure 13. Plot of the top two Principal components, 0 relates to no heart attack occurred; 1 relates to a heart attack occurred

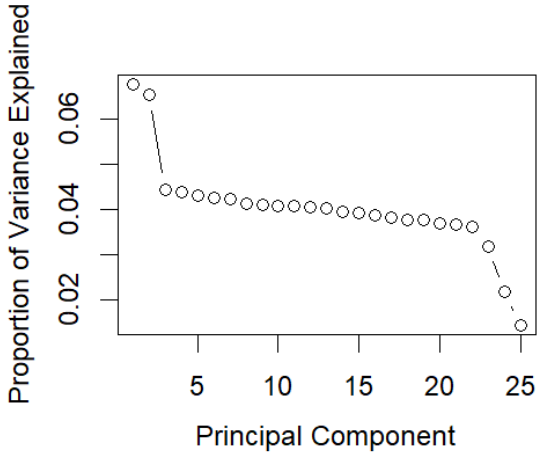

Figure 14. Twenty-five principal components and the variance they explain in the data.

PCA creates a principal component for each feature. Figure 11 shows the variance explained by each component. The top two components explain the most variance. Overall, PCA was not that successful at explaining all the variance due to the nature of the spread of the heart rate data set in feature space.

Logistic Regression

Logistic Regression models the relationship between a set of input features and the probability of a binary outcome using the logistic function. Given the binary nature of our response variable which is Heart Attack Risk, Logistic Regression was selected for its simplicity, interpretability, and suitability for binary classification tasks.

The initial Logistic Regression model was trained directly on the imbalanced dataset. The model predicted all cases as ”NoRisk,” leading to perfect recall (1.0) but zero specificity (Figure 23). Precision was 0.6526, and the Area Under the Curve (AUC) was 0.521, barely better than random guessing (0.5). The ROC curve, close to the diagonal line, further highlighted the model’s inability to distinguish between classes.

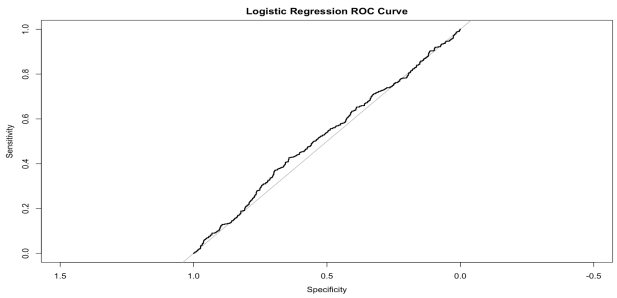

Figure 15. Logistic Regression ROC Plot

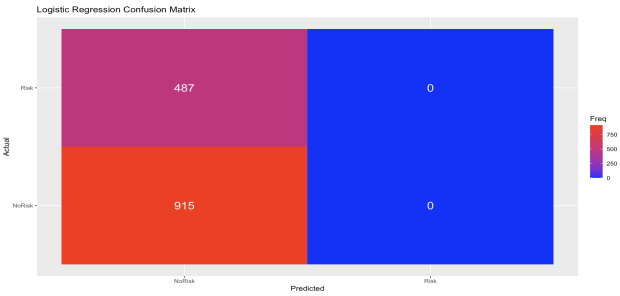

Figure 16. Logistic Regression Confusion Matrix

To address class imbalance, Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set. This balanced the class distribution, leading to more balanced predictions. Precision improved to 0.663, recall dropped to 0.767, and F1-Score reached 0.711 (Figure 23). The AUC remained low at 0.521, indicating limited discriminatory power despite balanced data.

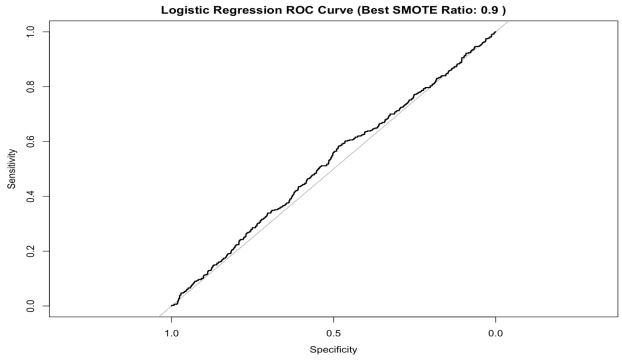

Figure 17. Logistic Regression SMOTE ROC Plot

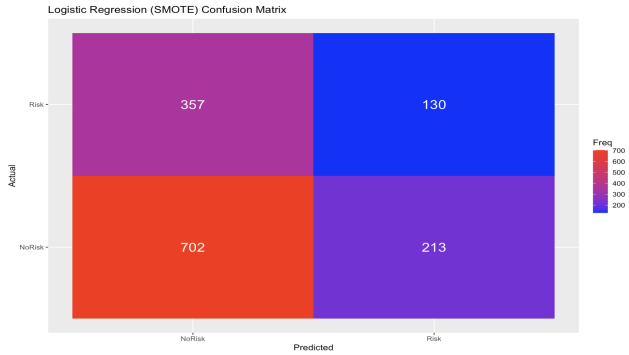

Figure 18. Logistic Regression SMOTE Confusion Matrix

The original Logistic Regression model achieved perfect recall at the expense of specificity, predicting almost all cases as ”NoRisk.” In contrast, the SMOTE-adjusted model provided a better balance, improving precision and achieving a more reasonable recall. However, the model’s AUC remained low, suggesting that even with balanced data, it struggled to distinguish between the two classes. This may be due to weak feature relevance or model limitations. Further improvements could include feature selection or regularization to enhance performance.

SVM

A support vector machine (SVM) is a supervised learning algorithm based on the idea of utilizing a hyperplane to divide the data in feature space. It attempts to find the maximum margin between the data sets.

SVM was chosen since it performs well on small- medium

sized data sets. There is also a lot of versatility in the types of data it can be applied to since it can adapt to both linear and non-linear distributions.

There are different kernel functions that can be utilized in SVM. The kernel function is the shape of the hyperplane. The commonly used kernel shapes are linear, polynomial, radial and sigmoid. We chose the polynomial kernel over radial, linear and sigmoid. The polynomial kernel resulted in the best accuracy and F1 score. This kernel allowed the model to find a maximum margin hyperplane that separates the data the best.

In Figure 17, the confusion matrix shows the number of true labels to predicted. The model predicted a majority of the inputs as a 0, no heart attack risk. This means that the hyperplane established by the model grouped a many of the true label 1s, heart attack risk present, with the other group. The lack of success with the SVM is a result of the limited distance in feature space present due to the high amount of overlap in the two classes. SVM is a model dependent on distance in feature space which struggles to separate data that is overlapping.

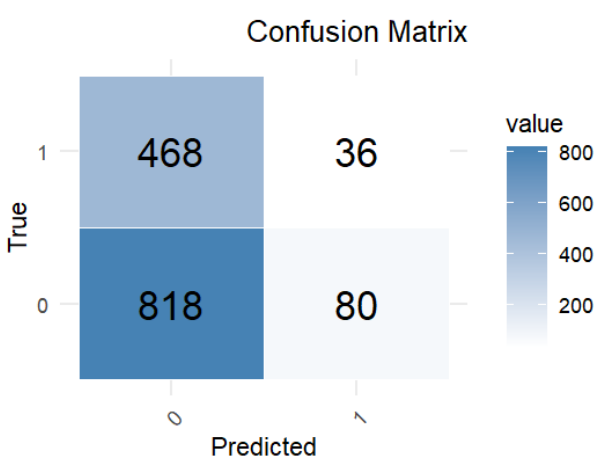

Figure 19. Support Vector Machine (SVM) predicted versus true labels

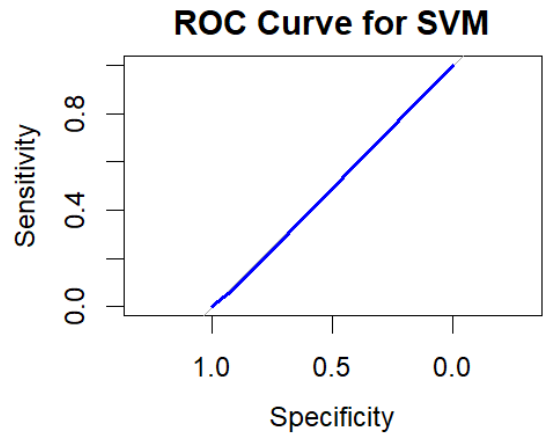

Figure 20. ROC curve for the SVM classifier

The ROC curve shows the tradeoff between sensitivity and specificity. A linear relationship between sensitivity and specificity shows an AUC of 0.5 meaning the model is performing at chance. The SVM model according to this metric is performing close to chance.

The model accuracy is 60.2 percent and F1 score is 0.776. SVM is not the best model choice for this data set. Since it is attempting to classify a real world problem the amount of noise and variability inherent to the dataset affecting the model performance.

KNN

K-Nearest Neighbors (KNN) is a non-parametric, instance-based learning algorithm that classifies observations based on the majority class among the K closest training samples. In this analysis, KNN was applied both on the original imbalanced dataset and after applying SMOTE to balance the class distribution.

In the initial application of KNN, the model was trained on the imbalanced dataset. An optimization process was used to determine the best value of K, with the AUC score declining rapidly as K increased. The confusion matrix reveals poor classification performance, with low specificity (0.3368) and a low balanced accuracy of 51.71%. This outcome indicates that the model struggles to distinguish between the two classes. The ROC curve confirms its weak discriminatory power, staying close to the diagonal.

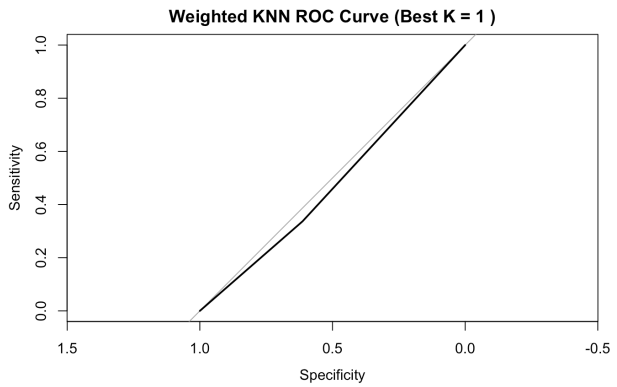

Figure 21. KNN ROC Plot

After SMOTE was applied, the KNN model was re-trained on the balanced dataset. The AUC values were optimized for the best K, but the model's performance remained weak. The confusion matrix showed slightly better distribution of predictions, but sensitivity and specificity remained low. The ROC curve remained close to the diagonal, highlighting the model's limited ability to separate the classes. Although SMOTE balanced the training data, KNN remained ineffective because the model is highly sensitive to distance, and the synthetic data may not reflect real-world patterns.

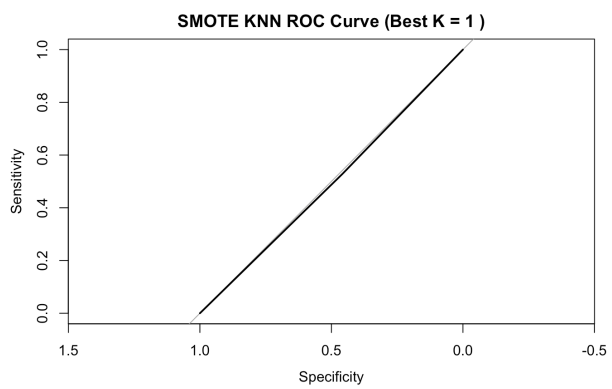

Figure 22. KNN SMOTE ROC Plot

KNN struggled both in its original form and with SMOTE. The model's reliance on distance makes it sensitive to feature scaling and synthetic data, which may not accurately reflect the true distribution of the classes. These limitations suggest that KNN may not be well-suited for this dataset, and alternative models or further optimization may be necessary.

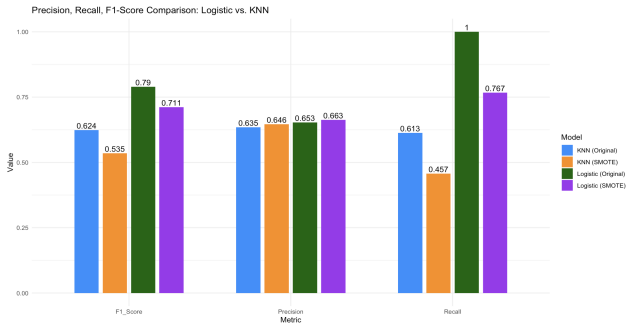

Figure 23. F1-Score, Precision, Recall, Comparison Plot

Naive Bayes

Additionally, we implemented a Naive Bayes classifier to predict heart attack risk. Naive Bayes is a probabilistic supervised learning algorithm that applies Bayes' theorem under the simplifying assumption that all features are conditionally independent given the class label. Despite this strong assumption, Naive Bayes is often highly effective in practice, especially for high-dimensional datasets, due to its simplicity, speed, and ability to handle categorical predictors directly.

The correlation heatmap (Figure 12) revealed that most predictor variables are weakly correlated with each other, supporting the core Naive Bayes assumption that the predictors are independent once we know the class label (i.e., the risk category). While some moderate correlations were observed — for example, between BMI and Obesity or Sex and Smoking — these do not substantially violate the model's assumptions, as Naive Bayes is known to be robust to mild dependencies among predictors. Given the categorical nature of several variables (e.g., Sex, Diet, Medication Use) and the interpretability and efficiency offered by Naive Bayes, it serves as a reasonable baseline model for this classification task. However, we remain mindful that stronger correlations between predictors may limit the model's ability to accurately combine evidence, and thus why we compare its performance to other classifiers.

For this model, we included both numeric and categorical predictors, ensuring that non-ordered categorical variables (such as country, hemisphere, continent, and diet) were

properly treated as factors rather than numeric codes. To handle the possibility of zero probabilities in the conditional likelihood estimates, we applied Laplace smoothing (with a factor of 1), ensuring stable probability calculations even for rare feature-category combinations.

After training the Naive Bayes model on the balanced dataset (post-SMOTE), we evaluated its performance on the held-out test set. The resulting confusion matrix in Figure 24 revealed an overall accuracy of 63.6%. More importantly, we computed the F1 Score for the positive (at-risk) class, which balances precision and recall to better reflect the model's ability to detect true positives under class imbalance. The Naive Bayes model achieved an F1 Score of 0.53, indicating moderate effectiveness in recovering true at-risk cases while balancing false positives.

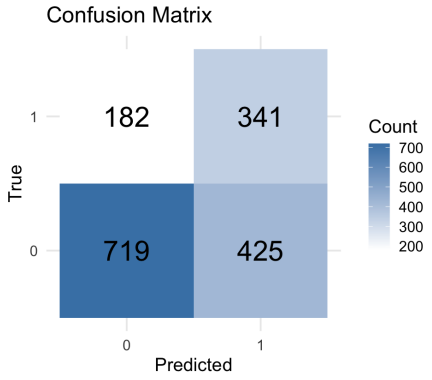

Figure 24. Confusion Matrix for Naive Bayes Model

The ROC curve (Figure 25) and corresponding AUC (Area Under the Curve) value provide a summary measure of the model's discriminative ability across all decision thresholds. The AUC of 0.65 suggests that the Naive Bayes classifier can rank a randomly selected at-risk individual above a randomly selected not-at-risk individual about 65% of the time — better than random guessing (AUC = 0.5).

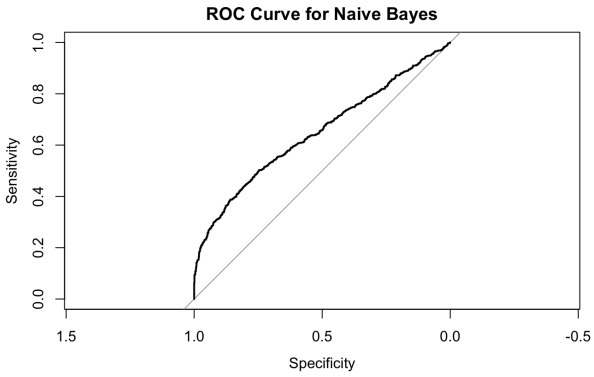

Figure 25. ROC Curve for Naive Bayes Model

Overall, while the Naive Bayes classifier offers advantages in simplicity and interpretability, its conditional independence assumption may limit its ability to capture the complex interactions between health and lifestyle variables present in heart attack risk prediction for this dataset. These results highlight the importance of selecting models that align well with the data's underlying structure for optimal predictive performance.

Random Forest

To model heart attack risk, we also applied a Random Forest classifier — a supervised ensemble learning method that combines the predictions of multiple decision trees to improve overall accuracy and reduce overfitting. Each tree in the forest is trained on a random subset of the data (bootstrapping) and considers a random subset of features at each split, ensuring model diversity and robustness. By aggregating the predictions from hundreds of individual trees through majority voting, the Random Forest effectively captures complex, nonlinear relationships between health, lifestyle, and demographic variables. Importantly, it also provides measures of feature importance, helping identify which predictors (such as BMI, cholesterol, sedentary time, and exercise) contribute most to distinguishing at-risk individuals.

In our implementation, we configured the Random Forest to build 500 trees, providing a large ensemble that improves stability and predictive performance by averaging across many independent learners. At each split within each tree,

the model randomly selected 5 variables (defined by the square root of the total number of predictors, as is standard for classification tasks) to reduce correlation between trees and prevent overfitting. These hyperparameter choices strike a balance between model complexity and generalization power, allowing the model to effectively capture the underlying patterns in the data.

After training the model on the balanced dataset (using SMOTE to correct for class imbalance), we evaluated its performance on the test dataset. The resulting confusion matrix is shown in Figure 26.

This produced an overall accuracy of 71.1%. However, since the problem involves an imbalanced dataset, we additionally calculated the F1 Score for the positive (at-risk) class, which balances precision and recall. The Random Forest achieved an F1 Score of 0.55, indicating moderate effectiveness in identifying true heart attack risk cases while managing false positives.

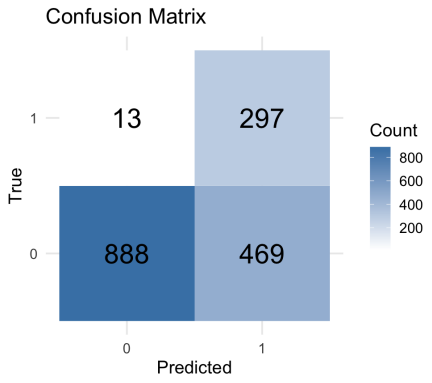

Figure 26. Confusion Matrix for Random Forest Model

The ROC curve was also plotted (see Figure 27.), and the calculated AUC (Area Under the Curve) provides a summary measure of the model's ability to distinguish between the two classes (at-risk vs. not-at-risk) across all possible classification thresholds. The AUC value of 0.723 indicates that the Random Forest model has reasonable discriminative power — meaning that, on average, it can correctly rank a randomly chosen positive (at-risk) case higher than a randomly chosen negative (not-at-risk) case about 72% of the time. While this is notably better than random guessing (AUC = 0.5), it also suggests there is room for improvement, especially for a sensitive application like heart attack risk prediction where high precision and recall are critical.

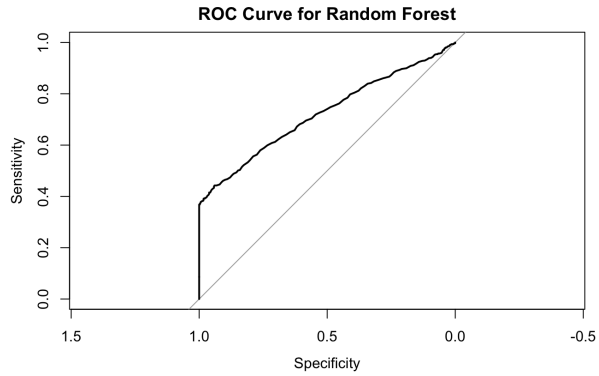

Figure 27. ROC Curve for Random Forest Model

Finally, the variable importance plot (see Figure 28) revealed which predictors contributed most strongly to the model's decisions. Features such as BMI, income, sedentary hours per day, and exercise hours per week ranked among the top contributors, highlighting important health and lifestyle factors associated with heart attack risk in the dataset.

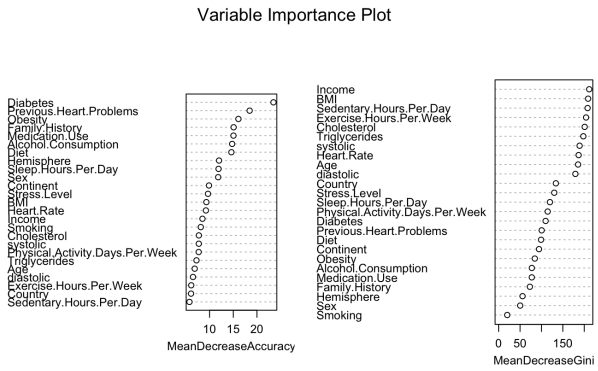

Figure 28. Variable Importance Plot for Random Forest Model

K-Means Clustering

To complement the supervised machine learning approaches, we also applied an unsupervised clustering method — K-means clustering — to explore whether the numeric features in the dataset naturally group individuals by heart attack risk without using labeled outcome information during training.

K-means is an iterative partitioning algorithm that divides the dataset into a predefined number of clusters (in this case, k=2), aiming to minimize within-cluster variance by assigning each observation to the nearest cluster centroid.

Importantly, for this analysis we excluded categorical predictors (such as sex, country, hemisphere, and diet) and focused solely on scaled numeric variables. This design choice was necessary because K-means relies on Euclidean distance to compute similarity between data points — a distance metric that is meaningful only for continuous numeric data. Including categorical variables without proper transformation (such as using specialized methods like K-prototypes) could have led to distorted or invalid clustering results, as the algorithm would mistakenly treat category codes as ordered numeric values.

After running the K-means algorithm with 25 random initializations to avoid local minima, we aligned the resulting cluster assignments to the true risk labels by examining the contingency table (cluster-to-label mapping) and selecting the mapping that maximized agreement. Despite this alignment, the K-means model produced an overall accuracy of only 53.3% on the dataset, indicating that the unsupervised clusters did not strongly correspond to the labeled heart attack risk categories.

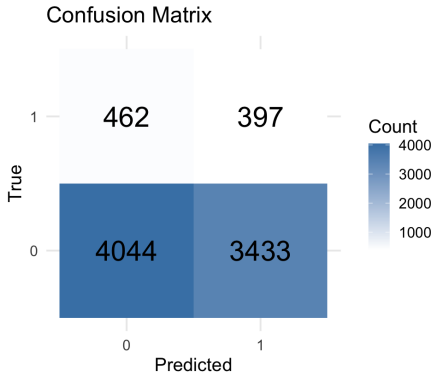

Figure 29. Variable Importance Plot for Random Forest Model

We further evaluated the performance using the F1 Score for the positive (at-risk) class, which accounts for both precision and recall. The resulting F1 Score was 0.17, reflecting poor ability to correctly identify at-risk individuals, largely due to the unsupervised nature of the method and thus the fact that the model does not know or use the true heart attack risk labels during training. That is, it grouped individuals solely based on feature similarity, not on their actual risk status.Additionally, the ROC curve (see Figure 30) and the calculated AUC (Area Under the Curve) were approximately 0.50, indicating no better than random discrimination between the two risk classes.

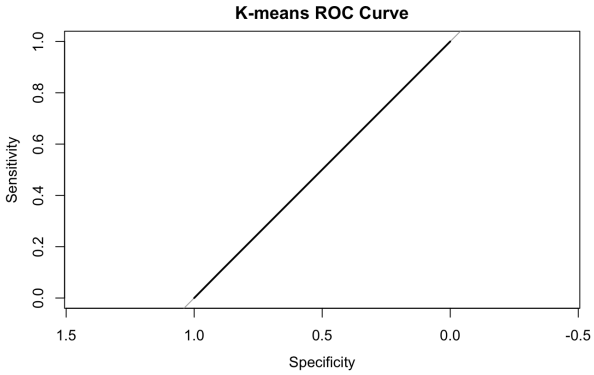

Figure 30. Variable Importance Plot for Random Forest Model

These results highlight the limitations of K-means clustering for this problem, suggesting that the underlying patterns separating at-risk and not-at-risk individuals are subtle and complex, requiring supervised learning models that can leverage labeled outcome data. Nonetheless, the K-means experiment provides valuable insight by illustrating that simple numeric clustering on the available features does not naturally recover the risk categories, reinforcing the importance of more sophisticated, supervised approaches like Random Forest for predictive modeling in this domain.

Conclusion

We applied six machine learning models to predict heart attack risk: Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Naive Bayes, Random Forest, and K-Means Clustering. Each model's performance was evaluated using accuracy and F1-Score as primary metrics (Figure 31).

Logistic Regression achieved 65.3% accuracy with an F1-Score of 0.79, showing strong performance due to its sim-

plicity and balanced predictions. SVM followed with 60.2% accuracy and an F1-Score of 0.78, leveraging its ability to handle non-linear data but slightly underperforming due to complex patterns.

Random Forest was the most accurate model (71.1%), but its F1-Score was lower (0.55), indicating high sensitivity but poor precision. Naive Bayes performed moderately, with 63.6% accuracy and an F1-Score of 0.53, limited by its assumption of feature independence.

KNN struggled with 51.7% accuracy and an F1-Score of 0.62, suffering from distance sensitivity and poor generalization even with SMOTE. K-Means Clustering, as an unsupervised method, showed the weakest performance, with 53.27% accuracy and an F1-Score of 0.17, unable to leverage class labels.

Overall, Random Forest and Logistic Regression emerged as the most effective models, with Random Forest offering the highest accuracy and Logistic Regression providing a balanced combination of precision and recall. SVM also demonstrated competitive performance, while KNN, Naive Bayes, and K-Means were less effective, largely due to their sensitivity to feature distributions and their limitations in capturing complex patterns within the data.

Model	Accuracy	F1 – Score
Logistic Regression	65.3%	0.79
Support Vector Machine (SVM)	60.2 %	0.78
K-Nearest Neighbor (KNN)	51.7%	0.62
Naive Bayes	63.6%	0.53
Random Forest	71.1%	0.55
K-Means Clustering	53.27%	0.17

Figure 31. Model Comparison

Discussion

While our best-performing model was only able to achieve an accuracy of approximately 71%, this result reflects the inherent complexity and variability present within human populations. Even when using features well-supported in the scientific literature as potential predictors of heart attack risk, there will inevitably be unexplained variance due to individual differences, measurement noise, and unobserved factors. Such discrepancies highlight the challenges of modeling real-world health outcomes, where predictive accuracy is limited not only by the quality of the data but also by the multifactorial nature of the condition.

Though SMOTE was applied during training to address class imbalance and help the model better learn patterns in the minority (at-risk) class, several factors can explain why performance across all classifiers used remains moderate. Although SMOTE balances the class distribution by generating synthetic examples based on nearest neighbors, it may not fully capture the true complexity or variability of real-world at-risk cases, which can limit generalization to unseen data.

Furthermore, heart attack risk is inherently driven by complex, multifactorial interactions — including genetic, behavioral, and environmental factors — many of which may not be fully captured in the available dataset. Even with strong ensemble methods like Random Forest, the presence of noisy or weakly predictive features can reduce the model's ability to sharply separate risk groups. Lastly, the presence of noisy or overlapping patterns between the classes may also contribute to misclassifications, particularly in borderline cases where feature values are similar for at-risk and not-at-risk individuals. As a result, while the overall accuracy remains relatively high due to correct majority class predictions, the F1 score — which balances precision and recall for the minority class — reflects the model's ongoing challenge in consistently and precisely identifying positive (at-risk) cases.

Looking forward, there are several avenues to potentially improve model performance. First, applying more advanced ensemble techniques such as Gradient Boosting Machines (GBM) or XGBoost could improve predictive accuracy, as these models often outperform Random Forest by optimizing errors sequentially and handling complex feature interactions with finer granularity. Additionally, incorporating hyperparameter tuning (using grid search or randomized search) could optimize the performance of models like SVM, KNN, and Random Forest beyond their default configurations.

Another important direction would be to improve the feature set itself. Incorporating additional clinical, genetic, or behavioral variables—if available—could capture more of the underlying factors influencing heart attack risk. More-

over, applying advanced feature engineering techniques may uncover hidden patterns missed by the original predictors.

Finally, we could explore cost-sensitive learning or weighted loss functions to better handle the practical importance of correctly identifying at-risk individuals, as well as advanced methods like stacking ensembles or neural networks to combine the strengths of multiple algorithms. These steps could help push the models beyond the performance plateau observed in this study and better meet the challenges of predicting complex health outcomes like heart attack risk.

References

1. Centers for Disease Control and Prevention. (n.d.). Heart disease facts. Centers for Disease Control and Prevention. https://www.cdc.gov/heart-disease/data-research/facts-stats/index.html

2. Benjamin, Emelia J., et al. "Heart Disease and Stroke Statistics—2019 Update: A Report from the American Heart Association." Circulation, vol. 139, no. 10, 5 Mar. 2019, pp. 56–528, www.ahajournals.org/doi/10.1161/CIR.0000000000000659.

3. Moraes-Silva IC, Rodrigues B, Coelho-Junior HJ, Jardim Feriani D, Irigoyen MC. Myocardial Infarction and Exercise Training: Evidence from Basic Science. Adv Exp Med Biol. 2017;999:139–53. [DOI] [PubMed]